AFN-1255 **PCA131**

P.G.D.C.A. DIPLOMA EXAMINATION, MAY 2011

First Semester

Computer Applications

FUNDAMENTALS OF DIGITAL COMPUTER

(Non-CBCS—2004 onwards)

Time : 3 Hours Maximum : 100 Marks

Part A (15 × 1 = 15)

Answer **all** questions.

1. What is a program ?

2. What is byte ?

3. What is ASCII ?

4. What is non-volatile memory ?

5. What is Register ?

6. Subtract the following Binary Numbers :–

$10101 \cdot 1010 - 10001 \cdot 0011.$

7. Write the switching postulates of Boolean Algebra.

8. Give the symbols for AND and OR Gates.

9. Define Bus.

10. What is Multi programming ?

11. What is Virtual Memory ?

12. What is DMA ?

13. Define the term “Source Program”.

14. What are the facilities provided by the OS ?

15. What are digital signal processors ?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

16. (a) Draw a block diagram of computer. Explain the function of each block.

(*Or*)

(b) Write a procedure to find the occurrence of a digit in the given sequence.

17. (a) What is the advantage of using Hexadecimal Numbers ?

(*Or*)

(b) Evaluate : (i) 11010 – 10111, (ii) 110 × 111.

18. (a) What is the difference between a Video graphic terminal and a Graph plotter ?

(*Or*)

(b) Explain the operation of a flip-flop.

19. (a) List the operation codes.

(*Or*)

(b) Show that $X \cdot Y + X \cdot Z + X \cdot Y \cdot Z = X \cdot Y + X \cdot Z$

20. (a) Explain how demand paging Algorithm works.

(*Or*)

(b) What do you understand by UNIX pipes ?

Part C (5 × 12 = 60)

Answer **all** questions.

21. (a) How will you represent integers and fractions ?

(*Or*)

(b) Explain input methods.

22. (a) Explain any two drives in detail.

(*Or*)

(b) How will you represent 2's Complement Numbers ? Explain.

23. (a) Explain the canonical forms of Boolean Functions.

(*Or*)

(b) Explain the Memory Communication.

24. (a) Write about high level languages.

(*Or*)

(b) Why do we need operating system ? Write about any two operating systems.

25. (a) Write about the Evolution of Micro computers.

(*Or*)

(b) Define Smart card. Classify the Generations of Computer.

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AFN-1256

PCA132

P.G.D.C.A. DIPLOMA EXAMINATION, MAY 2011

First Semester

Computer Applications

OFFICE AUTOMATION

(Non-CBCS—2004 onwards)

Time : 3 Hours Maximum : 100 Marks

Part A (15 × 1 = 15)

Answer **all** questions.

1. How will you restore windows ?

2. What is the process of arranging Icons ?

3. What is Switching of tasks ?

4. What is MS-word ?

5. Write the steps to Edit a document.

6. List the features of Mail Merge.

7. What is worksheet ?

8. How will you print a worksheet ?

9. Define Date and Time.

10. Write any *two* Mathematical functions.

11. How will you define the header and footer ?

12. List the steps to resize and moving a chart.

13. Define a Template.

14. Define Powerpoint views.

15. What is slide sorter view ?

Part B

(5 × 5 = 25)

Answer **all** questions.

16. (a) What is Menu ? List the operations in Menu.

(*Or*)

(b) What are the functions of program Manager ?

17. (a) Explain the usage of tabs and their types.

(*Or*)

(b) Explain the templates and wizards.

18. (a) List the steps in Editing the cells and functions.

(*Or*)

(b) List the steps in creating charts.

19. (a) Define Powerpoint. List the steps in creating a presentation.

(*Or*)

(b) List the steps in printing a a presentation.

20. (a) Write about formatting commands.

(*Or*)

(b) Write about Multiple worksheets.

Part C (5 × 12 = 60)

Answer **all** questions.

21. (a) List the steps in creating Multiple windows.

(*Or*)

(b) How will you run a MS-DOS program ?

22. (a) List the steps in :

(i) Finding

(ii) Replacing

(iii) Spell checking a text.

(*Or*)

(b) List the features of MS-word.

23. (a) List the functions of worksheet.

(*Or*)

(b) Write about the Addressing modes, Naming ranges.

24. (a) List the steps in creating a database in a worksheet.

(*Or*)

(b) Define Macros. List the steps in creating Macros.

25. (a) How will you view the powerpoint presentation ?

(*Or*)

(b) List the steps in printing a presentation.

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AFN-1257 **PCA133**

P.G. D.C.A. DIPLOMA EXAMINATION, MAY 2011

First Semester

Computer Applications

PROGRAMMING IN C

(Non-CBCS—2004 onwards)

Time : 3 Hours Maximum : 100 Marks

Part A (15 × 1 = 15)

Answer **all** questions.

1. What are the comments used in string ?

2. How will you represent a Bitwise operator XOR ?

3. Define an Unconditional statement.

4. Explain While loop.

5. What is indirection operator ?

6. Define an Array.

7. Strn cat (S1, S2, n). What does the function perform ?

8. How will you declare a function ?

9. What is a pointer ?

10. What is a structure ?

11. What is a Malloc ?

12. What are the advantages of using typdef in a program ?

13. What is a tree ?

14. What are sequential files ?

15. What are command line Arguments ?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

16. (a) Define Constants. Explain the types with relevant examples.

(*Or*)

(b) Explain about the following :

(i) Break.

(ii) Continue.

17. (a) How an array name is interpreted ? When it is passed to a function ?

(*Or*)

(b) Write a program to multiply any two Matrices.

18. (a) What is a stack ? How will you implement it ?

(*Or*)

(b) How a multi dimensional array defined in terms of a pointer ? Explain

19. (a) Write a factorial program to show the usage of Recursion.

(*Or*)

(b) Explain binary tree with an example.

20. (a) Write short notes on :

(i) fopen, fclose.

(ii) fread, fwrite.

(*Or*)

(b) Write about Macros in detail.

Part C

$(5 \times 12 = 60)$

Answer **all** questions.

21. (a) Explain conditional statements with suitable examples.

(*Or*)

(b) What are Bitwise operators ? How are they used ?

22. (a) What are multidimensional arrays ? Explain with an example.

(*Or*)

(b) Explain the string functions in detail.

23. (a) Distinguish between Structures and Unions.

(*Or*)

(b) Define List. Write a program to implement list.

24. (a) Explain Recursive Algorithms in detail.

(*Or*)

(b) Describe tree traversal Algorithm with an example.

25. (a) Write a C program to store text in a file and find the vowels present in the text.

(*Or*)

(b) Write a C program to implement Macros.

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AFN-1258 **PCA134**

P.G.D.C.A. DIPLOMA EXAMINATION, MAY 2011

First Semester

Computer Applications

VISUAL PROGRAMMING

(Non-CBCS—2004 onwards)

Time : 3 Hours Maximum : 100 Marks

Part A (15 × 1 = 15)

Answer **all** questions.

1. What is GUI ?

2. What are the different types of Brushes ?

3. Write about WinMain procedure.

4. What is the use of Text Boxes ?

5. Write down the properties of Window usage.

6. What is Single Document ? Interface.

7. When will you use picture box ?

8. Mention any two properties of Data Grid.

9. Write the function which will draw a circle.

10. What is the usage of CWnd function ?

11. Where will you Bitmaps ?

12. What is the use of Icon ?

13. What is DAO ?

14. Write about MFC.

15. Name any two Data Access methods.

Part B

(5 × 5 = 25)

Answer **all** questions.

16. (a) Write notes on Message Processing.

(*Or*)

(b) What is DLL ? Write its usage.

17. (a) Write notes on variables and constants in V.B.

(*Or*)

(b) What is Event procedure ?

18. (a) How will you use DDE Events, DDE methods ?

(*Or*)

(b) What is OLE ? Explain.

19. (a) What are the resources of MFC ?

(*Or*)

(b) Write a MFC program to display "WELCOME"

20. (a) Explain the advantages and disadvantages of DAO.

(*Or*)

(b) Write about database applications with multiple document usage.

Part C

(5 × 12 = 60)

Answer **all** questions.

21. (a) Write a SDK program for freehand drawing.

(*Or*)

(b) Explain the GDI functions.

22. (a) What are the various string functions in VB ?

(*Or*)

(b) Write notes on (i) Forms (ii) Single Document Interface.

23. (a) Explain about Activex control.

(*Or*)

(b) How to use Data control in V.B. application ? Explain with an example.

24. (a) Explain the AFX functions.

(*Or*)

(b) Write about ODBC.

25. (a) Write a MFC program to display a Bitmap.

(*Or*)

(b) Explain Dat Access Methods.

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AFN-1259 PCA231

P.G.D.C.A. DIPLOMA EXAMINATION, MAY 2011

Second Semester

Computer Applications

OBJECT ORIENTED PROGRAMMING AND C++

(Non-CBCS—2004 onwards)

Time : 3 Hours Maximum : 100 Marks

Part A (15 × 1 = 15)

Answer **all** questions.

1. Write notes on Software crisis.

2. Write down command structure of a C++ program.

3. What are command line arguments in C++?

4. What is friend function ?

5. Define the various parameter passing mechanisms supported by C++.

6. What is return by reference ?

7. How will you use arguments in C++ ?

8. What is the use of destructors ?

9. What is Binary operators ?

10. What is empty operators ?

11. What is abstract class ?

12. Write the difference of using private and protected member function.

13. Write about "try" statement.

14. What is generic class ?

15. Define Manipulators.

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

16. (a) Write a C++ program using inline function.

(*Or*)

(b) Explain about control structures in C++.

17. (a) How will you use two dimensional array in C++ ? Give an example.

(*Or*)

(b) What is Virtual function ?

18. (a) Explain Operator overloading with example.

(*Or*)

(b) Write about multiple constructors with an example.

19. (a) Describe the concept of parametrized constructors.

(*Or*)

(b) Write a C++ program to find whether the given word is palindrome or not ?

20. (a) Explain the file I/O functions.

(*Or*)

(b) What are generic classes and functions ?

Part C

(5 × 12 = 60)

Answer **all** questions.

21. (a) What is meant by token ? Explain the types of tokens with example.

(*Or*)

(b) Explain the control structures in C++.

22. (a) What is a class in C++ ? How does it differ from a structure in C ? Explain with example.

(*Or*)

(b) (i) Write a C++ program for friend function.

(ii) Write notes on pointers to members.

23. (a) What are read only objects ? What is the role of a constructor in such objects ?

(*Or*)

(b) Write about polymorphism.

24. (a) Explain the inheritance concept in detail.

(*Or*)

(b) How will you use nesting of classes ? Explain.

25. (a) Write about function template with an example.

(*Or*)

(b) Write about exception handling.

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AFN-1656 PCA232

P.G.D.C.A. DIPLOMA EXAMINATION, MAY 2011

Second Semester

Computer Applications

INTERNET PROGRAMMING

Time : 3 Hours Maximum : 100 Marks

Part A (15 × 1 = 15)

Answer **all** the questions.

1. What are the internet requirements ?

2. What is TCP / IP ?

3. Define IP address.

4. What do you mean by URL ?

5. Define Newsgroup.

6. What is Gopher ?

7. What is the use of 'What is' database ?

8. What is Telnet ?

9. Define IRC.

10. What declarations are required for every Java application ?

11. What are identifiers ?

12. What are calling methods ?

13. Differentiate between exception and error in Java ?

14. How will you concatenate two strings ?

15. What is I/O filter ?

Part B

(5 × 5 = 25)

Answer **all** questions.

16. (a) What is the nature of shell account?

(*Or*)

(b) Write short notes on – Online information services.

17. (a) Write notes on : Search Engines.

(*Or*)

(b) How to post your own articles in internet ?

18. (a) How does archie works ?

(*Or*)

(b) How mailing list is different from usenet news groups ?

19. (a) What are the operators used in Java ? Brief with example.

(*Or*)

(b) What is Package ? List the use of Package.

20. (a) What is the principle behind inheritance ? List out the advantages of inheritance.

(*Or*)

(b) What are Applet's life cycle methods ? Explain them.

Part C

(5 × 12 = 60)

Answer **all** the questions.

21. (a) (i) What do you need in order to use ISDN ? (6)

(ii) Write short notes on PPP account.

(*Or*)

(b) How to use mail from shell account ? Explain in detail.

22. (a) Explain in detail about URL schemes, host names and port names.

(*Or*)

(b) Explain Usenet.

23. (a) Explain the usage of FTP client for anonymous FTP.

(*Or*)

(b) Explain the types of internet talk facility.

24. (a) Explain the flow control statements in Java.

(*Or*)

(b) Brief the usuage of Interfaces with suitable example.

25. (a) Explain in detail about networking concept of Java language with example.

(*Or*)

(b) Describe AWT.

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P.G.D.C.A DIPLOMA EXAMINATION, MAY 2011

Second Semester

Computer Applications

R.D.B.M.S.

Time : 3 Hours Maximum : 100 Marks

Part A (15 × 1 = 15)

Answer **all** the questions.

1. What is file processing ?

2. What is attribute ?

3. Define data abstraction.

4. List the tools of Oracle.

5. Define Primary key.

6. How constraint is defined in table creation ?

7. What is NULL ?

8. What is the use of Ceil function ?

9. How user is created in SQL * PLUS ?

10. What is Object binding in Oracle ?

11. Give the syntax to create index.

12. Define View.

13. List any two privileges available in Oracle.

14. Who is DBA ?

15. How comments are represented in Report Writer ?

Part B (5 × 5 = 25)

Answer **all** the questions.

16. (a) Explain Codd's rule.

(*Or*)

(b) Describe Network model with example.

17. (a) Explain about DML statements.

(*Or*)

(b) Write short notes on Data types.

18. (a) Explain about Date functions.

(*Or*)

(b) Write short notes on Sub Queries.

19. (a) Write short notes on object definition in database.

(*Or*)

(b) Explain about views.

20. (a) Write short notes on Grant.

(*Or*)

(b) Explain in detail about commit.

Part C (5 × 12 = 60)

Answer **all** the questions.

21. (a) Explain in detail about Data Models.

(*Or*)

(b) Describe with sketch about DBMs architecture.

22. (a) (i) Write short notes on logical comparision. (6)

(ii) Explain Query expression Operator. (6)

(*Or*)

(b) Explain SQL statements.

23. (a) Explain about SQL functions.

(*Or*)

(b) Describe about Join Theory in SQL.

24. (a) Write short notes on :

(i) Sequence.

(ii) Synonyms.

(*Or*)

(b) Explain in detail about View.

25. (a) How DBA create users and grant privileges in SQL ? Explain with example.

(*Or*)

(b) Explain about Report writer of SQL.

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AFN-1658 PCA234

P.G.D.C.A. DIPLOMA EXAMINATION, MAY 2011

Second Semester

Computer Applications

SOFTWARE ENGINEERING

Time : 3 Hours Maximum : 100 Marks

Part A (15 × 1 = 15)

Answer **all** the questions.

1. List the Categories of Projects.

2. What do you mean by life cycle Model ?

3. How will you define solution strategy ?

4. Differentiate between design and architectural design.

5. Define Coupling.

6. What is Software metric model ?

7. What do you mean by faults ?

8. What is Reliability of models ?

9. What is the use of database as study tool ?

10. Define Testing Criteria.

11. What is psychology of testing ?

12. Define System testing.

13. What is data abstraction ?

14. Why ADA language is used in project definition ?

15. What is the syntax for defining exception Handling ?

Part B (5 × 5 = 25)

Answer **all** questions.

16. (a) Describe COCOMO.

(*Or*)

(b) Write short notes on team standards.

17. (a) Write short notes on Modularity Concepts.

(*Or*)

(b) Explain about Verification Methods.

18. (a) Write short notes on Software reliability.

(*Or*)

(b) Write short notes on availability models.

19. (a) Write short notes on Boundary value Analysis.

(*Or*)

(b) Explain about Mutation testing.

20. (a) Explain about Data abstraction.

(*Or*)

(b) Write short notes on Concurrency Models.

Part C

(5 × 12 = 60)

Answer **all** the questions.

21. (a) Explain about phased Life Cycle model.

(*Or*)

(b) Explain how you define a Software problem.

22. (a) Explain about design notations.

(*Or*)

(b) Explain in detail about Software Metrics.

23. (a) Explain the steps involved in preparing Manuals.

(*Or*)

(b) Explain about software errors, faults, repair and availability of software project.

24. (a) Explain about testing fundamentals.

(*Or*)

(b) Explain about the Metrics in Reliability estimation.

25. (a) Explain about ADA features and commands.

(*Or*)

(b) Describe in detail about Software development environments.

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